

Comparative Review of Mixed Reality and Virtual Reality: Concepts, Systems, and Application

Mr. Omkar Chikalge

Department of Computer Engineering.

Bharati Vidyapeeth's College Of Engineering, Lavale.

Pune, India.

omkarchikalge@gmail.com

Ms. Apurva Gawande

Department of Computer Engineering.

Bharati Vidyapeeth's College Of Engineering, Lavale.

Pune, India.

apurvagawande30@gmail.com

Prof. Yogesh Kadam

Department of Computer Engineering.

Bharati Vidyapeeth's College Of Engineering, Lavale.

Pune, India.

Yogesh.kadam@bharativedyapeeth.edu

Abstract

In today's era, the two closely related yet technically distinct technologies **Mixed Reality (MR)** and **Virtual Reality (VR)** have been emerged inside the domain of extended reality. In Virtual Reality, the user is placed inside a fully digital environment where the real surroundings are not visible. The process is simple it replaces the surroundings to 3D digital world by blocking out physical world using Head Mounted Display (HMD). On the other side MR is a mix of VR and AR, It integrates Virtual objects into the live view of Physical world and blends the two together. This Requires accurate spatial mapping, depth estimation, and real-time registration so that the virtual object is tied by the digital reference points into the physical reality and can't be blocked by any obstacles. Recent commercial devices such as Meta Quest 3 and Apple Vision Pro shows that how the high quality mixed reality advances in sensing, display and other aspects than that of the traditional VR models. This review summarizes core concepts, device architectures, software stacks, and interaction techniques for both mixed reality and virtual reality. It also discusses about the current application of MR and VR in multiple industrial areas such as Healthcare, Education, Construction, Architecture, etc. and also the challenges like Latency problem, visual quality, cost effectiveness, usability and

other human factors. Also conveys some future predictions where Artificial intelligence and advanced hardware will be integrating into the extended reality.

Keywords

Mixed reality, virtual reality, extended reality, spatial computing, head-mounted displays

1. Introduction

In today's era Virtual Reality (VR) and Mixed Reality (MR) have become important technologies inside extended reality. Virtual Reality replaces the real world completely into a digital environment, The process is simple it blocks the physical world using head mounted display and uses motion tracking so that the user feels present inside the virtual environment [1].

On the other side Mixed Reality allows both real and virtual objects to exist together and interact in real time. In MR the virtual objects are placed into the real world in such a way that they look fixed and can also be hidden behind real world objects [3][4].

With the development of video see through and optical see through headsets, Mixed Reality has moved from just a research concept to real world applications in industries, education and consumer usage [2].

Recent devices such as apple vision pro, microsoft hololens and meta quest 3 shows that how these technologies are improving, some devices provide full virtual experience while some provide mixed reality using cameras and sensors [2][6].

Understanding the difference between VR and MR is important for developers, researchers and industries so that they can choose the correct technology for applications like training, healthcare, industrial work and collaboration.

The main objective of this paper is to compare VR and MR based on concepts, system design, interaction, applications, challenges and future scope.

2. Literature Review

The existing research says that Virtual Reality is mainly used for creating fully immersive environments where the user is completely inside the digital world and the real world is completely blocked. It is usually used in gaming, simulations and training applications where full immersion is required [1].

On the other side Mixed Reality is used in situations where both real and virtual world are needed together. It allows virtual objects to be placed inside the real world and interact with it in real time. This makes it useful in industrial work, healthcare and education [3][7].

Many studies show that MR requires more advanced technologies compared to VR because it needs proper understanding of the real environment and it uses spatial mapping, plane detection, depth sensing and real time alignment to make virtual objects fixed in the real world [3].

Some research also explains that MR systems use technologies like SLAM (Simultaneous Localization and Mapping), computer vision and sensors to understand the environment [10].

From recent devices like Apple Vision Pro and Meta Quest 3 we learn that how mixed reality is improving using high quality cameras, sensors and AI for better environment understanding [2].

However research also states some limitations such as high cost of devices, latency issues, tracking errors and user discomfort which are still challenges in both VR and MR systems.

3. Proposed Methodology

I. Conceptual Background

A. Reality–Virtuality Continuum and Definitions

The reality–virtuality continuum is a range between real world and virtual world. Technologies like AR, MR and VR are in this range.

Virtual Reality (VR) completely replaces the real world with a fully digital environment using head mounted displays. It blocks the real world and gives full immersion to the user [1].

In Augmented Reality (AR) digital information is added on top of the real world but it does not properly interact with real objects.

Mixed Reality (MR) is more advanced where real and virtual objects exist together and interact with each other. In MR the virtual objects are placed in real world with proper alignment, lighting and they can also be hidden behind real objects [10]. The term Extended Reality (XR) is used as a common term for VR, AR and MR.

B. Core Interaction Characteristics

In Virtual Reality the user must feels present in virtual world so it uses wide view, low latency and high resolution displays while interaction is done using controllers or hand tracking [5].

Mixed Reality needs more than immersion. It needs proper understanding of the real environment hence it uses spatial mapping, plane detection and real time alignment so that virtual objects can interact with real world [2][5].

MR systems use technologies like computer vision, SLAM and depth sensing making them more complex than VR systems [10].

II. HARDWARE ARCHITECTURES

A. Displays and Optical Systems

VR headsets use dark displays like LCD or OLED which block the real world completely and show only virtual content.

MR systems use two types of approaches:

- Optical see through where virtual objects are projected into real world
- Video see through where cameras capture real world and combine it with virtual content

Devices like apple vision pro use micro OLED displays with very high resolution to provide both VR and MR experiences [2].

Many devices balance between resolution, field of view, cost and weight [6].

B. Sensing, Tracking and Passthrough

- Both VR and MR use tracking systems like 6 degree of freedom tracking using cameras and sensors.
- Inside out tracking is used where cameras and IMU sensors track the movement of head and hands.
- MR systems require additional sensors for depth sensing and environment understanding.
- Devices like apple vision pro use multiple cameras and sensors to create real time spatial mapping [2].
- Meta quest 3 also provides full colour passthrough for mixed reality.

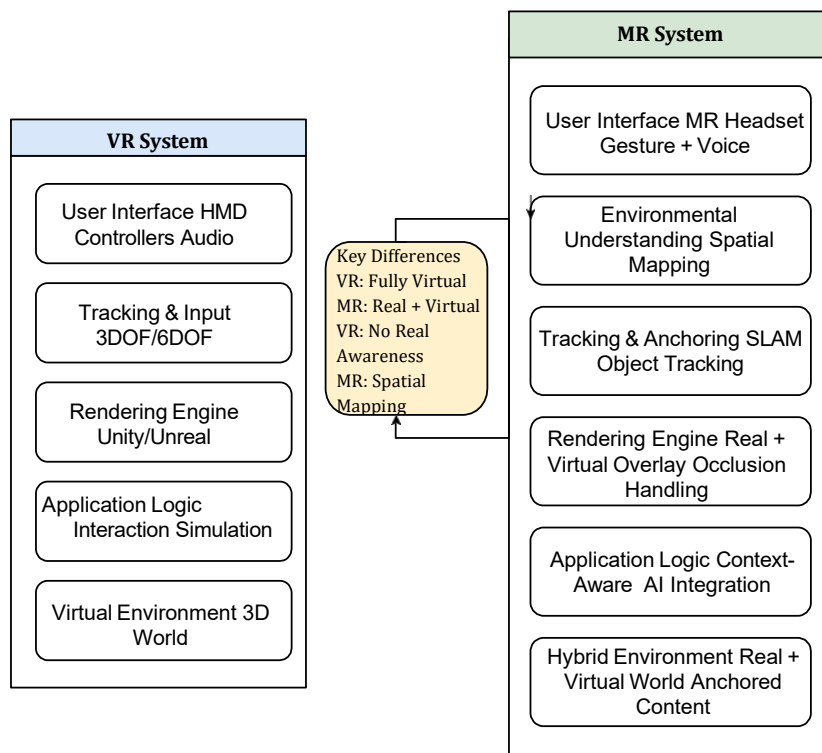
C. Compute, Power and Form Factor

Modern VR and MR devices use built in processors and AI accelerators so that they can work without external computers.

These devices require high processing power for rendering, AI and sensor data processing.

Because of this battery life, heat and device weight become important issues. But there is always a downgrading between performance, battery life and comfort.

D. Systematic Diagram of VR and MR



III. SOFTWARE STACKS AND INTERACTION MODELS

A. Operating Systems and SDKs

For development platforms like OpenXR, SteamVR and other SDKs are used by VR systems.

While, MR systems provide additional features like spatial mapping, pins or anchors and environment understanding using frameworks like mixed reality toolkit.

Apple's new model visionOS allows applications to be placed in 3D space and controlled using eyes, hands and voice [2][8].

B. Interaction Techniques

In VR interaction is mainly done using controllers with buttons and joysticks. In MR interaction is more natural using hand tracking, eye tracking and voice commands [8].

Eye tracking is also used for foveated rendering which improves performance. Advanced systems also use haptic devices like gloves to provide physical feedback. MR allows interaction with real objects along with virtual elements [8].

IV. APPLICATION DOMAINS

A. Training and Education

VR and MR are widely used in training where safety and cost are important. VR is used for simulations like medical training and industrial training. MR allows training in real environment with virtual instructions. It helps in improving learning and reducing training time [7][11].

B. Healthcare and Clinical Use

VR is used for medical simulations and training of doctors. MR is used for surgery guidance, patient education and treatment planning. It allows doctors to see 3D models along with real environment. Healthcare is one of the fastest growing areas for VR and MR [7].

C. Enterprise, Industrial and Collaborative Work

MR is used in industry for remote assistance, design review and maintenance. Workers can see instructions in real time while working. It helps in reducing errors and improving efficiency. Collaborative MR allows multiple users to work together in shared environment. VR is used for virtual meetings and simulations [9].

D. Entertainment and Consumer Applications

VR is mainly used in gaming and entertainment for immersive experiences. MR adds real world interaction into games. Users can interact with virtual objects in their own environment. Devices like meta quest 3 are improving mixed reality gaming.

4. Results and Discussion

A. COMPARATIVE ANALYSIS OF VR AND MR

Dimension	Virtual Reality (VR)	Mixed Reality (MR)
World Representation	Fully virtual environment	Real and virtual combined
Display	Opaque display	Optical or camera based
Environment Understanding	Limited understanding of surrounding and less spatial mapping	Advanced spatial mapping with focus on keeping environment intact
Interaction	Controllers, basic tracking	Hand, eye, voice interaction
Immersion vs Awareness	High immersion, low awareness	Balanced immersion and awareness
Devices	VR headsets	HoloLens, Vision Pro, Quest 3

B. Observations and Discussion

From the comparison it is observed that Virtual Reality completely replaces the real world and provides full immersion to the user. It blocks the physical environment and allows the user to interact only with the virtual environment [1].

Mixed Reality on the other side combines real and virtual world and allows interaction between both. Virtual objects are placed in real environment and they behave according to real world conditions. VR mainly focuses on immersion while MR focuses on interaction with real world [3].

In terms of applications:

- VR is used in gaming, simulations, virtual training and entertainment.
- MR is used in healthcare, industrial work, construction, design and real time guidance [7].

In VR systems the interaction is mostly done using controllers and head tracking. In MR systems interaction is more advanced using hand tracking, eye tracking and voice input.

Some challenges are also identified:

- Latency issues which affects performance
- Misalignment of virtual objects in MR
- High cost of MR devices
- Battery and power limitations
- User gets discomfort like eye strain and motion sickness

As MR needs to continuously understand the real environment it requires a lot of power [6].

In short VR is better for full immersive and gaming experience while MR is better for real world applications where interaction is needed [5][8].

5. Conclusion

From this study it is found out that Virtual Reality and Mixed Reality are both important technologies under extended reality but they work in different ways.

Virtual Reality creates a fully digital environment by blocking real world and focusing on virtual environment [1]. On the other hand Mixed Reality combines both real and virtual world and allows interaction with real environment [3].

From this study it is observed that VR is better for full immersive experiences like gaming and simulations while MR is more useful in real world applications like healthcare, industry and training where interaction with real environment is required [8][9][11].

This paper provides a clear comparison between Virtual Reality and Mixed Reality based on concepts, system design, applications and challenges. It helps in understanding the difference between both technologies and where they can be used.

This paper is based on theoretical analysis and existing research but no practical implementation is done. Also some limitations like cost, latency and hardware constraints are still present in both VR and MR systems.

6. Future Work

Challenges and Future Predictions on VR and MR

A. Latency and Visual Quality

- Both VR and MR require low latency for good user experience.
- In MR alignment errors between real and virtual objects can reduce performance.
- Camera quality, sensors and AI are important for improving MR quality.

B. Cost and Accessibility

- MR devices are expensive and not easily accessible.
- VR devices are cheaper but still need improvement in usability.
- Content creation for MR is also difficult and requires skills.

C. Human Factors and Safety

- Users may experience motion sickness, eye strain and discomfort.
- MR also has safety risks because users move in real world while using devices.
- Privacy is also a concern because MR devices capture real world data.

D. Future Outlook

- Future devices will combine VR and MR into one system.
- AI will improve scene understanding and interaction [9].
- Hardware will become lighter, faster and more efficient.

- MR will be used more in education, industry and collaboration.
- VR, AR and MR will play important role in metaverse systems in future [11].

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